

MATH 15200: Calculus II

Homework 2: Properties of the Integral

Due: Monday, October 12, 2026, at 11:59 PM.

This is the written portion of Homework 2; complete the routine online portion separately in Edfinity. Upload your written solutions to Gradescope, via Canvas. You may collaborate, but you must write and submit your own solutions independently.

Recall: Let f be continuous on an open interval I , and fix c in I . By the Fundamental Theorem of Calculus, the function $F(x) = \int_c^x f(t) dt$ is differentiable on I , with $F'(x) = f(x)$; and if $G' = f$ on I , then $\int_u^v f(t) dt = G(v) - G(u)$ for all u and v in I . Moreover, for all u , v , and w in I ,

$$\int_u^w f(t) dt = \int_u^v f(t) dt + \int_v^w f(t) dt \quad \text{and} \quad \int_v^u f(t) dt = - \int_u^v f(t) dt.$$

1. Let f be continuous on an open interval I , and let a and b be differentiable functions whose values lie in I . Prove that

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t) dt = f(b(x)) b'(x) - f(a(x)) a'(x).$$

Hint: Fix c in I and remember that $\int_{a(x)}^{b(x)} f(t) dt = \int_{a(x)}^c f(t) dt + \int_c^{b(x)} f(t) dt$. Let $F(u) = \int_c^u f(t) dt$. First show that $\frac{d}{dx} \int_c^{b(x)} f(t) dt = f(b(x)) b'(x)$ by writing this integral as $F(b(x))$ and applying the chain rule. Do a similar procedure with $\int_{a(x)}^c f(t) dt$.

2. Suppose that $a < b$, that f and g are continuous on $[a, b]$, and that $f(x) \leq g(x)$ for all x in $[a, b]$. Prove that

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

Hint: Compare Riemann sums for f and g that use the same partition of $[a, b]$ and the same sample points. Then take a limit, recalling that limits preserve non-strict inequalities ($\leq$).

3. The function $\sqrt{1+x^4}$ has no antiderivative given by a formula in familiar functions, so the integral below cannot be evaluated directly with the Fundamental Theorem of Calculus. Use Problem 2 to show that

$$1 \leq \int_0^1 \sqrt{1+x^4} dx \leq \frac{11}{10}.$$

Hint: For the upper bound, show that $\sqrt{1+x^4} \leq 1 + \frac{1}{2}x^4$ by expanding $(1 + \frac{1}{2}x^4)^2$ and comparing it with $1 + x^4$; explain why comparing squares is enough. Then integrate $1 + \frac{1}{2}x^4$ over $[0, 1]$.